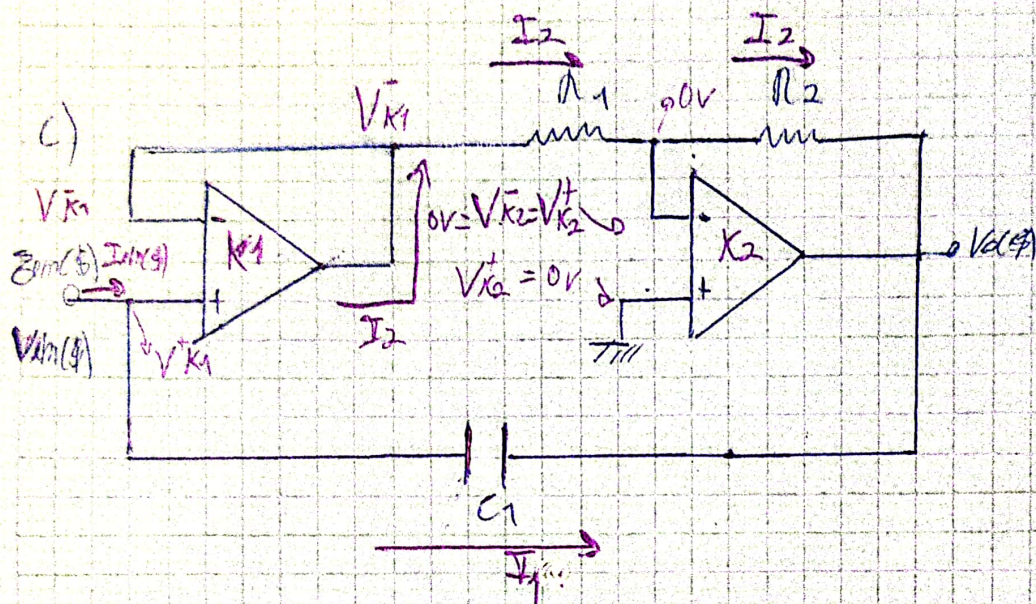




- $V^- = V^+$ por equi-
potencialidad
en K_1 y K_2 .
- aproximación ideal

$$Z_{in}(s) = \frac{V_{in}(s)}{I_{in}(s)}$$

$$Z_c(s) = \frac{1}{sC}$$

$$Z_a(s) = \infty$$

$$I_{in}(s) = \frac{V_{in}(s) - V_O(s)}{Z_c(s)} = I_1(s)$$

$$\text{con } V_{K1}^+ = V_{K1}^- = V_{in}(s) \text{ y } V_{K2}^+ = V_{K2}^- = 0V$$

$$I_2(s) = \frac{V_{K1}^- - V_{K2}^-}{Z_{R1}(s)} = \frac{V_{K1}^- - V_O(s)}{Z_{R2}(s)} \Rightarrow \frac{V_{in}(s) - 0V}{R1} = \frac{0V - V_O(s)}{R2}$$

$$\text{luego } \Rightarrow I_1(s) = \frac{V_{in}(s) - V_O(s)}{1/sC1} \quad \text{y reemplazando } \Rightarrow -\frac{V_{in}(s) \cdot R2}{R1} = V_O(s) \quad (*)$$

$$\text{En } (*) \Rightarrow I_{in}(s) = I(s) = sC1 \cdot \left[V_{in}(s) - (-) \frac{V_{in}(s) \cdot R2}{R1} \right]$$

$$\Rightarrow I_{in}(s) = V_{in}(s) \cdot sC1 \cdot \left[1 + \frac{R2}{R1} \right]$$

$$\Rightarrow I_{in}(s) = V_{in}(s) \cdot sC1 \cdot \left(\frac{R1 + R2}{R1} \right)$$

$$\text{Finalmente } \Rightarrow Z_{in}(s) = \frac{V_{in}(s)}{I_{in}(s)} = \frac{V_{in}(s)}{V_{in}(s) \cdot sC1 \cdot \left(\frac{R1 + R2}{R1} \right)} = \frac{R1}{sC1(R1 + R2)}$$